



Monitoring Report

CARBON OFFSET UNIT (CoU) PROJECT



Title: “15.65 MW Bundled Wind Energy Project in India by Intech Group”

Version 1.1

Date:08.01.2026

First CoU Issuance Period: 04 years and 04 months.

Monitoring Period: 01/09/2020 to 31/12/2024 (first and last date included)

8 DECENT WORK AND
ECONOMIC GROWTH



13 CLIMATE
ACTION



7 AFFORDABLE AND
CLEAN ENERGY





Monitoring Report (MR) CARBON OFFSET UNIT (CoU) PROJECT

Monitoring Report	
Title of the project activity	“15.65 MW Bundled Wind Energy Project in India by Intech Group”
UCR Project Registration Number	539
Version	1.1
Completion date of the MR	08/01/2026
Monitoring period number and duration of this monitoring period	Monitoring Period Number: 1.1 Duration of this monitoring Period: 01/09/2020 to 31/12/2024 (first and last date included)
Project participants	Inno Automation, Intech Automation, Intech Power Kovai, Intech Power Chennai Pvt Ltd.
Host Party	INDIA
Applied methodologies and standardized baselines	ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources - Version 22.0
Sectoral scopes	01 Energy industries (Renewable/Non-renewable Sources)
Annual amount of GHG emission reductions for this monitoring period in the registered PCN	2020: 1,961 CoUs (1,961 tCO ₂ eq)
	2021: 24,466 CoUs (24,466 tCO ₂ eq)
	2022: 14,840 CoUs (14,840 tCO ₂ eq)
	2023: 16,518 CoUs (16,518 tCO ₂ eq)
	2024: 11,376 CoUs (11,376 tCO ₂ eq)
Total:	69,161 CoUs (69,161 tCO ₂ eq)

SECTION A. Description of project activity

A.1. Purpose and general description of project activity >>

a) Purpose of the project activity and the measures taken for GHG emission reductions >>

Purpose of the Project Activity:

The project activity titled “15.65 MW Bundled Wind Energy Project in India by Intech Group” is spread across villages of Sakri in Maharashtra & villages in Bhelahadi Karnataka state in India.

The details of the registered project are as follows:

b) Brief description of the installed technology and equipment>>

The wind bundle Sr. No 01 is owned by Inno Automation. (Project Proponent or PP) Having the total installed capacity is of 2.5 MW and consists of 2 Wind turbine generators (WTGs) each of 1.25 MW capacity in Dhule district of Maharashtra. The wind bundle Sr. No 02 is owned by Intech Automation – (Project Proponent or PP) The total installed capacity is 2.5 MW wind power project in dhule of Maharashtra. PP consists of 2 WTGs of 1.25 MW each. The wind bundle Sr. No 03 Intech Power Kovai is the total installed capacity is 1.25 MW and consists of one WTG in Dhule district of Maharashtra.

The wind farm bundle Sr. No 04 is owned by Intech Power Chennai pvt ltd – (Project Proponent or PP) having 2 states in that Maharashtra & Karnataka in which the total capacity is 9.4 MW. Maharashtra having 6.25 MW & Karnataka 3.15 MW.

The bundled wind power projects, owned by Inno Automation, Intech Automation, Intech Power Kovai, and Intech Power Chennai Pvt Ltd, aim to generate clean electricity through renewable wind energy, displacing fossil fuel-based power generation and reducing anthropogenic greenhouse gas (GHG) emissions. Project activity consists of 24 WEGs (1.25) kw (MH) & 0.225 kw (KA) of Suzlon make S-66. The project involves the supply, erection, commissioning, and operation of these machines, managed by Suzlon. The wind farm generates Estimated approximately **31,532 MWh** of clean electricity annually, which is supplied to the state electricity utility MSEDCL & KPTCL. Addressing the energy demand-supply gap in Maharashtra & karnataka and supporting the region’s sustainable growth.

The primary objective of the project is to produce emission-free, environmentally friendly electricity by harnessing the wind energy potential in the region. By displacing fossil fuel-based power generation, the project reduces estimated greenhouse gas (GHG) emissions by approximately **(27,026 tCO₂eq)** tonnes of CO₂ equivalent (tCO₂e) while also mitigating other pollutants such as SO_x and NO_x. In the absence of this project, the equivalent electricity would have been generated by predominantly fossil fuel-based power plants in the Indian grid, which represents the pre-project scenario.

During the current monitoring period i.e 01/09/2020 to 31/12/2024, the project activity has generated 79,236.97 Mwh of clean electricity and reduced greenhouse gas (GHG) emissions by tonnes of CO₂ equivalent 69,161 (tCO₂eq).

Bundled Sr No	Name Of Wind Farm Bundle	Installed Capacity	Village	District	State
---------------	--------------------------	--------------------	---------	----------	-------

		(Mw)			
1	Inno Automation	2.5	Sakri	Dhule	Maharashtra
2	Intech Automation	2.5	Sakri	Dhule	Maharashtra
3	Intech Power Kovai	1.25	Sakri	Dhule	Maharashtra
4	Intech Power Chennai Pvt Ltd.	6.25	Sakri	Dhule	Maharashtra
5	Intech Power Chennai Pvt Ltd.	3.15	Bhelahadi	Gadag	Karnataka

Business Transfer Agreements and Power Purchase Agreement for Wind Energy Projects

On September 1, 2020, Inno Automation India Pvt Ltd entered into Business Transfer Agreements (BTAs) with two sellers: Palms Enercon Inc (Seller Location: G006) and Chawala Brothers Pvt Ltd (Seller Location: K295) for the acquisition of their respective business assets. Substation for Maharashtra turbine Jamde/220kv

Additionally, on August 1, 2020, Intech Automation Private Limited signed a BTA with Kajaria Ceramics Limited (Seller Locations: K103 and K104) for a 2.50 MW project.

Furthermore, a Power Purchase Agreement (PPA) was executed between Maharashtra Electricity Distribution Company Limited for a 6.25 MW capacity, covering projects at multiple locations: Project Location K132, J 095, J 096, J 097, J 098 Tal. Sakri, Dist. Dhule, Maharashtra.

A Business Transfer Agreement (BTA) for Turbine No. G1 and G31 was signed between M/s BSC Textiles and M/s Intech Chennai Pvt. Ltd. on June 7, 2019. Substation for Karnataka site Gadag 110/33kv

Turbines No. G3, G4, G7, G8, G11, G12, G32, G33, and G34 are associated with M/s B.S. Channabasappa & Sons and M/s Intech Chennai Pvt. Ltd. June 7, 2019.

Turbine No. G19 is associated with M/s SK Shashidhar and M/s Intech Chennai Systems Pvt. Ltd. June 7, 2019.

Turbine No. G10 is associated with M/s Shree Channeshwara Drier and M/s Intech Chennai Systems Pvt. Ltd. June 7, 2019.

Please refer section B.1 for installed technology and equipment.

c) Relevant dates for the project activity (e.g. construction, commissioning, continued operation periods, etc.)>>

The first WEC under the project activity was commissioned on 30.06.2005 and the last WEC under the project activity was commissioned on 13.08.2008. The commissioning date for all the WECs included in the project activity is given on the table below:

Customer Name	Substation	No Of WEC	Capacity	Commissioning Dates	
				HTSC No.	
Inno Automation India Pvt Ltd	Jamde/220kv	1	1.25	K295	21.03.2006
Inno Automation India Pvt Ltd	Jamde/220kv	1	1.25	G06	31.12.2005
Intech Automation Pvt Ltd	Jamde/220kv	1	1.25	K103	29.09.2005
Intech Automation Pvt Ltd	Jamde/220kv	1	1.25	K104	30.09.2005
Intech Power Kovai Pvt Ltd	Jamde/220kv	1	1.25	K294	21.03.2006
Intech Systems Chennai Pvt Ltd	Jamde/220kv	1	1.25	K132	30.06.2005
Intech Systems Chennai Pvt Ltd	Jamde/220kv	1	1.25	J095	29.09.2006
Intech Systems Chennai Pvt Ltd	Jamde/220kv	1	1.25	J096	29.09.2006
Intech Systems Chennai Pvt Ltd	Jamde/220kv	1	1.25	J097	29.09.2006
Intech Systems Chennai Pvt Ltd	Jamde/220kv	1	1.25	J098	29.09.2006
Intech Systems Chennai Pvt Ltd	Gadag 110/33kv	1	0.225	IGT 01	21.02.2008
Intech Systems Chennai Pvt Ltd	Gadag 110/33kv	1	0.225	IGT 02	21.02.2008
Intech Systems Chennai Pvt Ltd	Gadag 110/33kv	3	0.675	IGT 03, 04 & 08	21.02.2008
Intech Systems Chennai Pvt Ltd	Gadag 110/33kv	1	0.225	IGT 05	21.02.2008
Intech Systems Chennai Pvt Ltd	Gadag 110/33kv	1	0.225	IGT 07	21.02.2008.
Intech Systems Chennai Pvt Ltd	Gadag 110/33kv	1	0.225	IGT 10	21.02.2008
Intech Systems Chennai Pvt Ltd	Gadag 110/33kv	1	0.225	IGT 11	21.02.2008
Intech Systems Chennai Pvt Ltd	Gadag 110/33kv	1	0.225	IGT 12	21.02.2008
Intech Systems Chennai Pvt	Gadag	1	0.225	IGT	13.08.2008

Ltd	110/33kv			31	
Intech Systems Chennai Pvt Ltd	Gadag 110/33kv	2	0.45	IGT 32 & 34	21.02.2008
Intech Systems Chennai Pvt Ltd	Gadag 110/33kv	1	0.225	IGT 33	21.02.2008

First CoU Issuance Period: 04 years and 04 months.

UCR Project ID or Date of Authorization: 539

Start Date of Crediting Period: 01/09/2020 to 31/12/2024

Project Commissioning: 30.06.2005

d) Total GHG emission reductions achieved or net anthropogenic GHG removals by sinks achieved in this monitoring period>>

The total GHG emission reductions achieved in this monitoring period is as follows:

Summary of the Project Activity and ERs Generated for the Monitoring Period	
Start date of this Monitoring Period	01/09/2020
Carbon credits claimed up to	31/12/2024
Total ERs generated (tCO _{2eq})	69,161 tCO _{2eq}
Leakage	0

e) Baseline Scenario>>

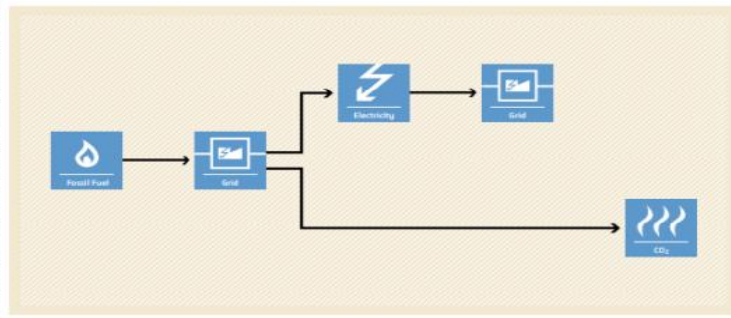
The baseline scenario identified at the PCN stage of the project activity is:

The scenario existing prior to the implementation of the project activity, is electricity delivered to the facility by the project activity that would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources. This is a green field project activity. There was no activity at the site of the project participant prior to the implementation of this project activity. Hence pre-project scenario and baseline scenario are the same.

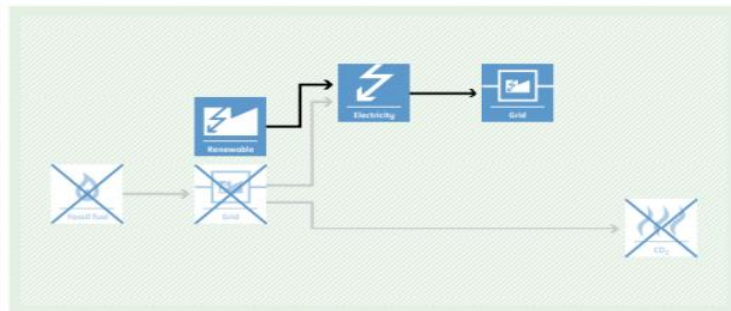
As per the approved consolidated methodology ACM0002 Version 22.0, if the project activity is the installation of a new grid-connected renewable power plant/unit, the baseline scenario is the following: “If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources to the grid”.

The Schematic diagram showing the baseline scenario:

BASELINE SCENARIO
Electricity provided to the grid by more-GHG-intensive means.



PROJECT SCENARIO
Electricity is generated and supplied to the grid using renewable energy technologies.



A.2. Location of project activity>>

Country: India

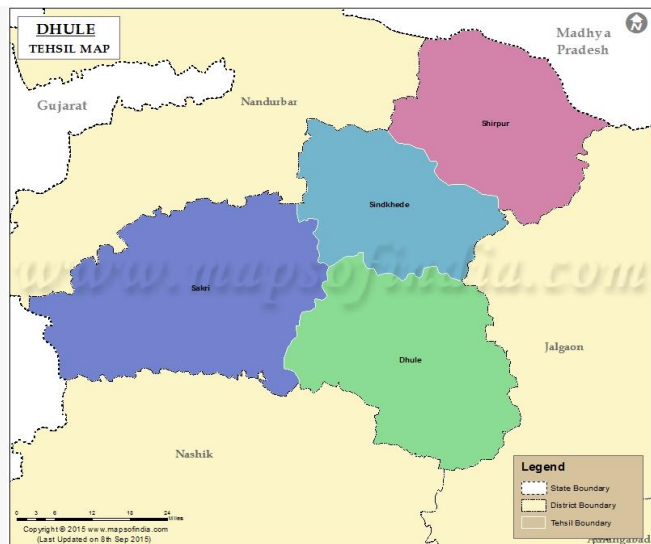
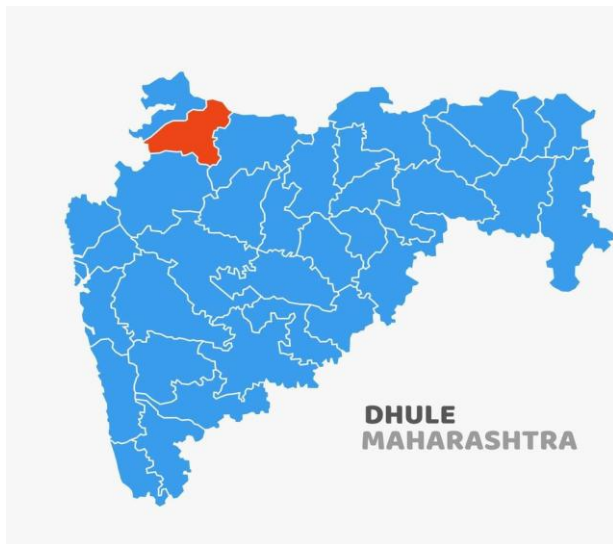
District: (MH) Dhule, (KA) Belahandi

Village: (MH) – Sakri (KA) – Gadag

State: Maharashtra, Karnataka



KARNATAKA



MAHARASHTRA

The Project consists of 24 no turbines of S-66 WECs of 1.25 kW. The Latitude & Longitude of the Project Activity Are Given Below.

Customer Name	Capacity	Site	Capacity		Geo Coordinates
			HTSC No.	Total Mw	
Inno Automation India Pvt Ltd	1.25	Sakri	K295	15.65	21°11'37.6"N, 74°12'30.7"E
Inno Automation India Pvt Ltd	1.25	Sangli	G06		17°10'04.6"N, 74°54'50.3"E
Intech Automation Pvt Ltd	1.25	Sakri	K103		21°13'04.6"N, 74°19'35.1"E
Intech Automation Pvt Ltd	1.25	Sakri	K104		21°13'12.1"N, 74°19'52.9"E
Intech Power Kovai Pvt Ltd	1.25	Sakri	K294		21°11'55.5"N, 74°12'48.1"E
Intech Systems Chennai Pvt Ltd	1.25	Sakri	K132		21°13'57.9"N, 74°24'02.0"E
Intech Systems Chennai Pvt Ltd	1.25	Sakri	J095		21°14'14.5"N, 74°24'02.2"E
Intech Systems Chennai Pvt Ltd	1.25	Sakri	J096		21°14'01.8"N, 74°24'21.0"E
Intech Systems Chennai Pvt Ltd	1.25	Sakri	J097		21°14'25.4"N, 74°24'33.3"E
Intech Systems Chennai Pvt Ltd	1.25	Sakri	J098		21°11'54.5"N, 74°17'53.4"E
Intech Systems Chennai Pvt Ltd	0.225	Gadag	IGT 01		15°20'10.3"N, 75°36'58.7"E
Intech Systems Chennai Pvt Ltd	0.225	Gadag	IGT 02		15°20'12.3"N, 75°36'52.4"E

Intech Systems Chennai Pvt Ltd	0.675	Gadag	IGT 03, 04 & 08		15°20'06.8"N ,75°36'52.8"E
Intech Systems Chennai Pvt Ltd	0.225	Gadag	IGT 05		15°20'09.6"N ,75°36'45.9"E
Intech Systems Chennai Pvt Ltd	0.225	Gadag	IGT 07		15°20'15.9"N ,75°36'45.8"E
Intech Systems Chennai Pvt Ltd	0.225	Gadag	IGT 10		15°20'12.5"N ,75°36'38.1"E
Intech Systems Chennai Pvt Ltd	0.225	Gadag	IGT 11		15°20'07.3"N ,75°36'38.6"E
Intech Systems Chennai Pvt Ltd	0.225	Gadag	IGT 12		15°20'04.6"N ,75°36'31.1"E
Intech Systems Chennai Pvt Ltd	0.225	Gadag	IGT 31		15°20'15.4"N ,75°36'23.5"E
Intech Systems Chennai Pvt Ltd	0.45	Gadag	IGT 32 & 34		15°20'12.4"N ,75°36'17.5"E
Intech Systems Chennai Pvt Ltd	0.225	Gadag	IGT 33		15°19'46.2"N ,75°36'19.0"E

A.3. Parties and project participants >>

Party (Host)	Participants
INDIA	PP: Inno Automation, Intech Automation, Intech Power Kovai, Intech Power Chennai Pvt Ltd. Aggregator: www.viviidgreen.com Viviid Emissions Reductions Universal Private Limited Address: 1001-B, Sri Krishna Complex, New Link Road, Opp. Laxmi Industrial Estate, Andheri West, Maharashtra 400053

A.4. References to methodologies and standardized baselines >>

SECTORAL SCOPE – 01 Energy industries (Renewable/Non-renewable sources)

TYPE: Renewable Energy Projects

CATEGORY- ACM0002., Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0.

A.5. Crediting period of project activity >>

The start date of the crediting period under UCR is considered from 01/09/2020

A.6. Contact information of responsible persons/entities >>

Name: Lokesh Jain
Contact: 9667647537
Email: lokesh.jain@viviidgreen.com
Address: 1001-B, Sri Krishna Complex, New Link Road, Opp. Laxmi Industrial Estate, Andheri West, Maharashtra 400053 .

SECTION B. Implementation of project activity

B.1. Description of implemented registered project activity >>

a) Provide information on the implementation status of the project activity during this monitoring period in accordance with UCR PCN>>

The project consists of four bundled wind farms with a total installed capacity of 15.65 MW, utilizing wind turbine generators (WTGs) to convert wind energy into electricity. These WECs are fully operational and are supplying electricity to the (MH, KA) Energy Transmission Corporation

The (WECs) were commissioned in phases across villages, including Sakri and Ghatnandre in Maharashtra, as well as Gadag in Karnataka. The first WEC under the project was commissioned on 30.06.2005 and the last was commissioned on 13.08.2008.

During the current monitoring period, all installed WEGs continue to operate as planned. The electricity generated is consistently fed into the state electricity grid through established interconnections with the state utility, supporting renewable energy generation, contributing to grid stability and displacing equivalent GHG emissions.

b) For the description of the installed technology(ies), technical process and equipment, include diagrams, where appropriate>>

The project consists of four bundled wind farms with a total installed capacity of 15.65 MW, utilizing wind turbine generators (WTGs) to convert wind energy into electricity. The details of the installed technology and equipment are as follows:

The 15.65 MW wind power project involves 10 Turbines S-66 Wind Energy Converters (WECs), each with a 1.25 -kW capacity, & KA having 14 turbines which has a capacity of 0.255 KW set up by Suzlon in India. These wind turbines convert wind energy into electricity using synchronous generators, which are manufactured at Suzlon's is spread across villages of Maharashtra in Dhule district & Karnataka in Gadag. Plant using advanced vacuum impregnation technology for better insulation and durability. The turbines have rotor blades, a nacelle with the generator and control systems, a tower, and a concrete foundation. The electricity generated is fed into the Indian grid through transformers. Without this project, the same amount of electricity would come from fossil fuel-based power plants, which is the baseline scenario. This renewable energy project reduces emissions and supports local manufacturing through technology transfer.

MODEL NO S-66

POWER	
Rated power:	1.25 kW
Flexible power ratings:	-
Cut-in wind speed:	4.0 m/s
Rated wind speed:	12.0 m/s
Cut-out wind speed:	20.0 m/s

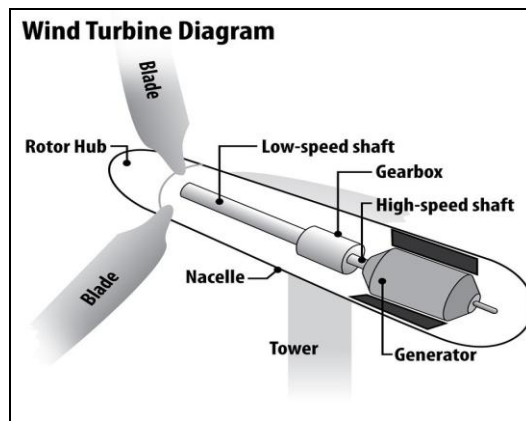
Survival wind speed:	52.5 m/s
Wind zone (DIBt):	-
Wind class (IEC):	-

ROTOR	
Diameter:	66.0 m
Swept area:	3,421.0 m ²
Number of blades:	3
Rotor speed, max:	20.6 U/min
Tipspeed:	71 m/s
Type:	-
Material:	GFK / Epoxy
Manufacturer:	-
Power density 1:	365.4 W/m ²
Power density 2:	2.7 m ² /kW

Gear box	
Type:	spur/planetary
Stages:	3
Ratio:	0.09375
Manufacturer:	-

Generator	
Type:	Asynchronus
Number:	1
Speed, max:	1,506.0 U/min
Voltage:	690.0 V
Grid connection:	50.0 Hz
Manufacturer:	-

Tower	
Hub height:	74.5 m
Type:	Steel tube
Shape:	conical
Corrosion protection:	painted
Manufacturer:	-



Main component of the windmill is explained below:

Main Tower:

This is a very tall structure with a door and inside ladder at the bottom. The door is used to enter into the tower for operation and maintenance.

Blades:

The windmills are provided with three blades. The blades are self-supporting in nature made up of Fibre Reinforced Polyester. The blades are mounted on the hub.

Nacelle:

The Nacelle is the one which contains all the major parts of a windmill. The nacelle is made up of thick rugged steel and mounted on a heavy slewing ring. Under normal operating conditions, the nacelle would be facing the upstream wind direction.

Hub:

The Hub is an intermediate assembly between the wing and the main shaft of the wind turbine. Inside the hub, a system to actuate the aerodynamic brake is fitted. The hub is covered with nose cone.

Main Shaft:

The shaft connects the gear box and the hub. Solid high carbon steel bars or cylinders are used as main shaft. The shaft is supported by two bearings

B.2 Do no harm or Impact test of the project activity>>

There are social, environmental, economic and technological benefits which contribute to sustainable development.

- **Social benefits:**

- The project activity will contribute to socio-economic development through improving the infrastructure for road network and other mode of communications in the remote part of the state during both the construction and operational period.
- The project activity will utilize renewable energy source for electricity generation instead of fossil fuel-based electricity generation which would have emitted gaseous, liquid and/or solid effluents/wastes. Thus, the project causes no negative impact on the surrounding environment and contributes to environmental well-being.
- The project activity will contribute towards reduction of the GHG emissions as well as emission of pollutants like SO_x, Suspended Particulate Matters (SPMs) etc. by avoiding equivalent amount of power generation from fossil fuel-based power plants.

- **Environmental benefits:**

- Utilizing wind energy instead of burning fossil fuels for electricity generation significantly decreases the emission of harmful pollutants, fostering cleaner air, water, and soil.
- Leveraging wind energy aids in preserving natural resources and minimizing detrimental impacts on the environment, contributing to overall ecological well-being.
- Moreover, harnessing wind energy offers a sustainable alternative to burning fossil fuels, which not only mitigates pollution but also conserves natural habitats and biodiversity, supporting healthier ecosystems and enhancing environmental resilience.

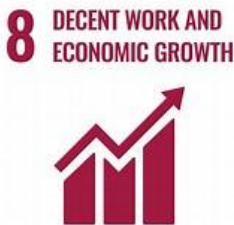
- **Economic benefits:**

- The project will generate electricity utilizing renewable source like wind, thus will increase the contribution of renewable based power generation in the region and will also help in reducing the demand - supply gap of the respective grid.
- The project activity involves substantial amount of investment, thus will contribute towards generation of direct and indirect employment opportunities as per the requirement of the skilled and semi-skilled manpower.
- Use of a renewable source of energy reduces the dependence on imported fossil fuels and associated price variation, thereby leading to increased energy security.

Positive contribution of the project to the following

Sustainable Development Goals:

- **SDG13:** Climate Action
- **SDG 7:** Affordable and Clean Energy
- **SDG 8:** Decent Work and Economic Growth

Development Goals	Targeted SDG	Target Indicator (SDG Indicator)
SDG 13: Climate Action 	13.2: Integrate climate change measures into national policies, strategies and planning Target	13.2.1: Number of countries that have communicated establishment or operationalization of an integrated policy/ strategy/ plan which increases their ability to adapt to the adverse impacts of climate change, and foster climate resilience and low greenhouse gas emissions development in a manner that does not threaten food production (including a national adaptation plan, nationally determined contribution, national communication, biennial update report or other)
SDG 7: Affordable and Clean Energy 	7.2: By 2030, increase substantially the share of renewable energy in the global energy mix Target	7.2.1: Renewable energy share in the total final energy consumption.
SDG 8: Decent Work and Economic Growth. 	8.5: By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value Target: Training, O&M staff.	8.5.1: Average hourly earnings of female and male employees, by occupation, age and persons with disabilities. No of employee: 07

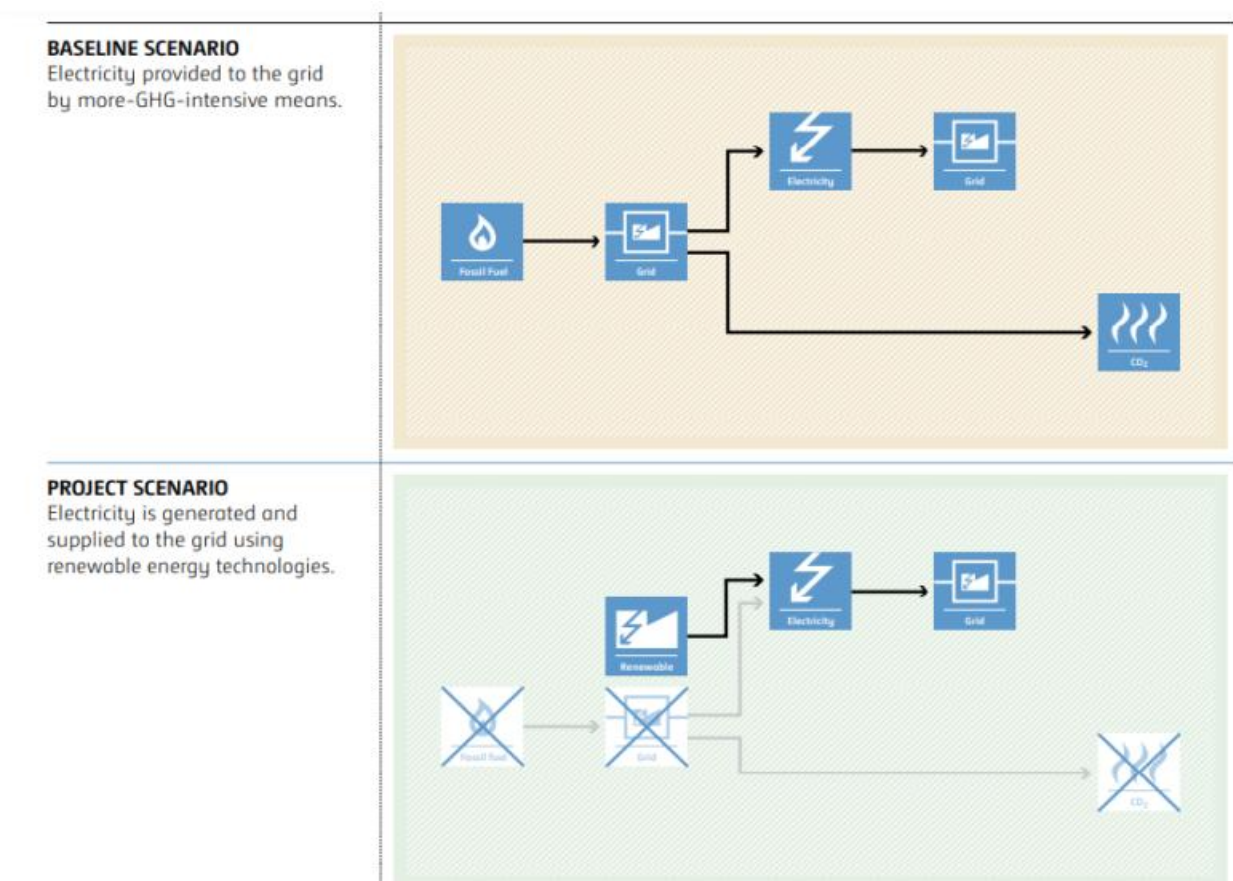
B.3. Baseline Emissions>>

The baseline scenario identified at the PCN stage of the project activity is:

The scenario existing prior to the implementation of the project activity, is electricity delivered to the facility by the project activity that would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources. This is a green field project activity. There was no activity at the site of the project participant prior to the implementation of this project activity. Hence pre-project scenario and baseline scenario are the same.

As per the approved consolidated methodology ACM0002 Version 22.0, if the project activity is the installation of a new grid-connected renewable power plant/unit, the baseline scenario is the following: “If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources to the grid”.

The Schematic diagram showing the baseline scenario:



B.4. Debundling>>

This Project is not a debundled component of a larger project activity.

SECTION C. Application of methodologies and standardized baselines

C.1. References to methodologies and standardized baselines >>

SECTORAL SCOPE – 01 Energy industries (Renewable/Non-renewable sources)

TYPE - Renewable Energy Projects

CATEGORY - ACM0002., Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0

C.2. Applicability of methodologies and standardized baselines >>

Applicability Criteria.	Applicability status
1) This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s), or (e) Involve a replacement of (an) existing plant(s)/unit(s). (f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	The proposed project involves establishing a new grid-connected renewable wind power plant, confirming to the specified criteria.
2) In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s). (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is located at site of the greenfield renewable power plant.	The project entails installing a new grid-connected renewable wind power project without the integration of a Battery Energy Storage System (BESS). Therefore, this condition does not apply to the project activity.
3) The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum	The proposed project involves installing new wind power plants without integrating a Battery Energy Storage System (BESS). Thus, the mentioned criterion does not apply

historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies² may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period. (e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode	
4) In case of hydro power plants, one of the following conditions shall apply: a) The project activity is implemented in an existing single or multiple reservoirs, with no change in the volume of any of reservoirs; or b) The project activity is implemented in an existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density calculated using equation (7) is greater than 4 W/m²; or c) The project activity results in new single or multiple reservoirs and the power density calculate equation (7), is greater than 4 W/m². d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density of any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply. (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²;	The proposed project involves the installation of wind power plants/units. Hence, the mentioned criterion is not applicable.

(ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity. (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² are: a) Lower than or equal to 15 MW; and b) Less than 10 per cent of the total installed capacity of integrated hydro power project.	
5) In the case of integrated hydro power projects, project proponent shall: a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability indifferent seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum five years prior to implementation of CDM project activity.	The proposed project activity involves the installation of wind power plants/units. Therefore, the mentioned criteria are not applicable.
6) In the case of PSP, the project participants shall demonstrate in the PDD that the project is not using water which would have been used to generate electricity in the baseline.	The proposed project activity involves installing wind power plants/units. Therefore, the specified criteria are not applicable.
7) The methodology is not applicable to: a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; b) Biomass-fired power plants;	The proposed project activity involves installing wind power plants/units. Therefore, the specified criteria are not applicable.
8) In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance	The proposed project activity involves installing wind power plants/units. Therefore, the specified criteria are not applicable.

C.3 Applicability of double counting emission reductions >>

There is no double accounting of emission reductions in the project activity due to the following reasons:

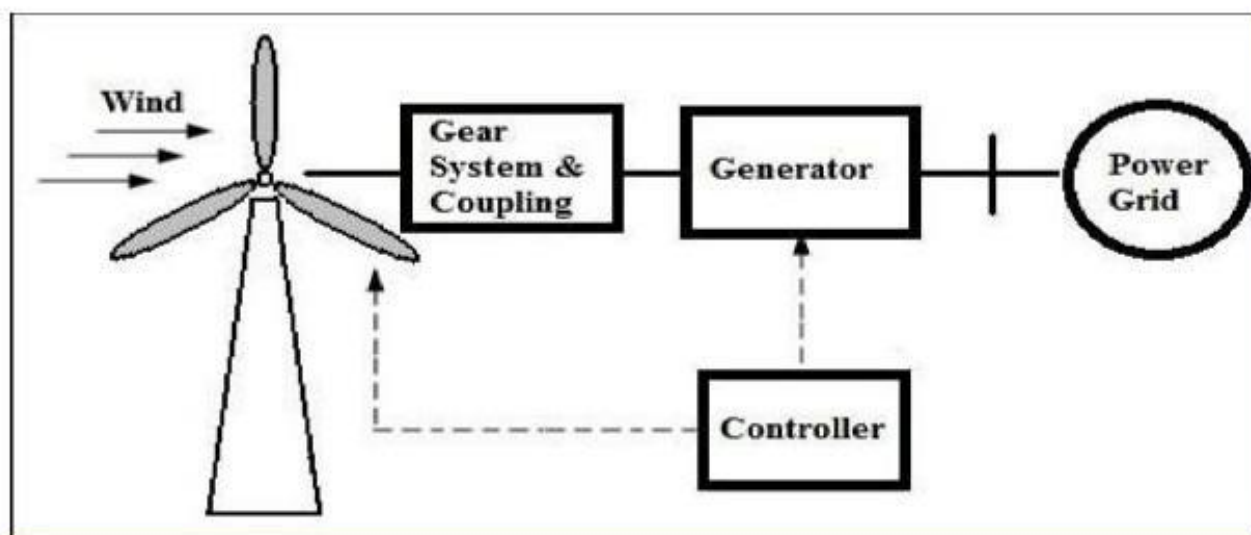
- Project is uniquely identifiable based on its location coordinates,
- Project has dedicated commissioning certificate and connection point,
- Project is associated with energy meters which are dedicated to the consumption point for project developer.

C.4. Project boundary, sources and greenhouse gases (GHGs)>>

According to the methodology ACM0002, version 22.0.0 the spatial extent of the project boundary includes the project power plant, and all power plants connected physically to the electricity system that the project power plant is connected to.

The project boundary includes the WECs of the project activity, transformer, individual meters, substation & Indian which is final consumer of generated electricity.

A schematic of project boundary diagram is shown below.



Project boundary:

The baseline study of the Indian grid shows that the main sources of GHG emissions under the baseline scenario are CO₂ emissions from the conventional power generating systems. Other emissions are that of CH₄ and N₂O but both emissions have been excluded for simplification.

The project activity generates.

Source	GHGs	Included?	Justification/Explanation
--------	------	-----------	---------------------------

Baseline scenario	Grid connected electricity generation	CO ₂	Yes	In the baseline scenario, the electricity would have been sourced from the Indian grid which in turn would be connected to fossil fuel fired power plants which emit CO ₂ .
		CH ₄	No	No methane is expected to be emitted.
		N ₂ O	No	No nitrous oxide is expected to be emitted.
Project Scenario	Greenfield wind energy conversion system	CO ₂	No	The project activity does not emit any emissions.
		CH ₄	No	No methane is expected to be emitted.
		N ₂ O	No	No nitrous oxide is expected to be emitted.

C.5. Establishment and description of baseline scenario (UCR Protocol) >>

As per the approved consolidated methodology ACM0002. version – 22.0, if the project activity is the installation of a new grid-connected renewable power plant/unit, the baseline scenario is the following:

“The baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise, been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid”

The project activity involves setting up of a new grid connected wind power plant to harness the green power from wind energy. In the absence of the project activity, the equivalent amount of power would have been supplied by the Indian grid, which is fed mainly by fossil fuel fired plants. The power produced at grid from the other conventional sources which are predominantly fossil fuel based. Hence, the baseline for the project activity is the equivalent amount of power produced at the Indian grid.

As per approved consolidated methodology ACM0002, version 22.0, emission reduction is estimated as difference between the baseline emission and project emission after factoring into leakage

Emission reductions are calculated as per methodology ACM0002, Version 22.0 Equation 17:

$$ER_y = BE_y - PE_y \quad (\text{Eq. 1})$$

Where,

ER_y = Emissions reductions in year y (t CO₂)

BE_y = Baseline emissions in year y (t CO₂)

PE_y = Project emissions in year y (t CO₂)

Baseline Emissions

The baseline emissions as per methodology ACM0002, Version 22.0, para 57 ; encompass solely the CO₂ emissions stemming from electricity generation in power plants displaced by the project activity. The methodology operates on the assumption that any electricity generation exceeding baseline levels would have originated from established grid-connected power plants and the integration of new grid-connected power plants.

The Baseline emissions as per methodology ACM0002, Version 22.0 Equation 11 in year y can be calculated as follows:

$$BE_y = EG_{PJ,y} * EF_{grid,y}$$

Where:

BE_y = Baseline emissions in year y (tCO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)

$EF_{grid,y}$ = Grid Emission factor in year y (tCO₂/MWh)

Since the project activity is the installation of a new grid connected renewable power plant (green field project), hence, $EG_{PJ,y}$ has been calculated as:

$$EG_{PJ,y} = EG_{facility,y}$$

Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

A "grid emission factor" denotes the CO₂ emission factor (measured in tCO₂/MWh) associated with each unit of electricity supplied by an electricity system. The UCR suggests employing an emission factor of 0.9¹ from 2013 to 2023 and Emission Factor of 0.757 tCO₂/MWh for 2024 as a cautious estimate for Indian projects. The same emission factor is utilized for computing emission reductions for the Project Activity.

Project Emission:

Regarding project emissions, ACM0002 version 22.0 specifies that only emissions related to fossil fuel combustion, emissions from the operation of geothermal power plants due to the release of non-condensable gases, and emissions from water reservoirs of hydroelectric plants should be taken into

¹As per [UCR CoU Standard Update: 2024 Vintage UCR Indian Grid Emission Factor Announced | by Universal Carbon Registry | Jan, 2025 | Medium](#)

account. Since the project involves a wind power project, emissions from renewable energy plants are negligible
Hence (PE_y = 0).

Leakage Emission:

Leakage, as outlined in ACM0002 version 22.0, para 5.6, is considered to be zero as there is no transfer of energy-generating equipment in the project activity

While the actual emission reduction achieved during the initial crediting period will be submitted during the first monitoring and verification, an ex-ante estimation is provided for reference.

Sr. No.	Year	Capacity	Total EGy ,Net Generation	Emission Factor	Baseline Emissions (BE)	Project Emissions (PE)	Emission Reductions
		MW	kWh	tCO ₂ /mWh	tCO ₂ e	tCO ₂ e	tCO ₂ e
1	01-06-2020 to 31-12-2020	15.65	2179934.372	0.9	1961	0	1961
2	01-01-2021 to 31-12-2021		27185237.02	0.9	24466	0	24466
3	01-01-2022 to 31-12-2022		16489277.44	0.9	14840	0	14840
4	01-01-2023 to 31-12-2023		18353956.34	0.9	16518	0	16518
5	01-01-2024 to 31-12-2024		15028566.4	0.757	11376	0	11376
	Total		79236971.56		69161		69,161

MWh = 79236.97

In Karnataka, net generation has been calculated by applying a 115% adjustment to the imported power recorded by the energy meter, in accordance with the regulations of the Karnataka Electricity Regulatory Commission. In Maharashtra, no such provision or requirement exists; therefore, no adjustment has been applied in the net generation calculation.

Total baseline emission reductions (BE_y) = 69,161 CoUs (69,161 tCO₂eq)

C.6. Prior History>>

The project activity has not been registered or applied for voluntary carbon benefits under any other GHG program and hence there is no double counting issue of CoUs.

C.7. Monitoring period number and duration>>

First Monitoring Period: 04 years, 04 months

Period: 01/09/2020 to 31/12/2024 (inclusive of both dates).

C.8. Changes to start date of crediting period >>

In accordance with the date stipulated in the Business Transfer Agreements (BTA) of the project proponents, the start date of crediting under UCR has been considered from 01/09/2020.

C.9. Permanent changes from PCN monitoring plan, applied methodology or applied standardized baseline >>

There are no permanent changes from registered PCN monitoring plan and applied methodology

C.10. Monitoring plan>>

Data and Parameters available ex-post verification

Data/Parameter	$EG_{pj,y}$
Data unit	MWh
Description	Net electricity supplied to the grid by the Project activity.
Measurement methods and procedures	Data Type: Measured Monitoring equipment: Energy Meters are used for monitoring Calibration frequency: once in five years (as per CEA Indian provision) Cross checking: Quantity of net electricity supplied to or consumed at PP's facility will be cross-checked from the monthly bills or invoices raised. The Net electricity supplied to the grid will be calculated by the values of electricity export to the grid. The Net electricity is recorded as follows: Thus, $EG_{PJ,y} = EG_{Net,Export}$
Value Applied	79236.97 Mwh
Monitoring frequency	The net energy exported to the grid is measured every month using calibrated energy meter by the State Electricity Board authorities in the presence of the project implementer or its representatives. The meter/s shall be jointly inspected and sealed by authorised

	representatives of the company and the state utility. Measuring procedure: Will be measured by an export-import energy meter. The net electricity exported by the project plant would either be directly sourced as a measured parameter or be calculated by deducting the amount of imported electricity from the total amount of exported electricity. Accuracy class of energy meter: 0.2s Calibration Frequency: As per the Central Electricity Authority the testing and calibration frequency should be once in five years².
Purpose of data	For baseline emission calculations

Data and Parameters available at validation (ex-ante values):

Data / Parameter:	EF _{grid,y}
Data unit:	tCO ₂ /MWh
Description:	A "grid emission factor" refers to a CO ₂ emission factor (tCO ₂ /MWh) which will be associated with each unit of electricity provided by an electricity system. The UCR recommends an emission factor of 0.9 tCO ₂ /MWh for the period 2013 - 2023 and 0.757 tCO ₂ /MWh from 2024 as a fairly conservative estimate for Indian projects. Hence, the same emission factor has been considered to calculate the emission reduction under conservative approach.
Source of data:	UCR CoU Standard Update: 2024 Vintage UCR Indian Grid Emission Factor Announced by Universal Carbon Registry Jan, 2025 Medium
Measurement procedures (if any):	-
Monitoring frequency:	Ex-ante fixed parameter
QA/QC procedures:	For the calculation of Emission Factor of the grid
Any comment:	-

Appendix: 1 Calibration of meters

² https://cea.nic.in/wp-content/uploads/2020/02/meter_reg.pdf

CALIBRATION DATA							
Sr.no	Meter Type	Meter Sr. No.	Sub-station	Accuracy class	Make	Calibration Details	Calibration Validity
K295	Main Meter	14796470	Valve/220kv	0.2	ELSTER ALPHA A1800	21.10.2023	20.10.2028
	Check meter	14796436	Valve/220kv	0.2	ELSTER ALPHA A1800	21.10.2023	20.10.2028
K294	Main meter	14796470	Valve/220kv	0.2	L&T	31.07.2024	31.06.2029
	Check meter	14796436	Valve/220kv	0.2	ELSTER ALPHA A1800	21.10.2023	20.10.2028
k104	Main meter	2839067	Jamde/220kv	0.2	ELSTER ALPHA A1800	04.06.2024	03.06.2029
	check meter	2839066	Jamde/220kv	0.2	ELSTER ALPHA A1800	04.06.2024	03.06.2029
k103	Main meter	2839067	Jamde/220kv	0.2	ELSTER ALPHA A1800	04.06.2024	03.06.2029
	check meter	2839066	Jamde/220kv	0.2	ELSTER ALPHA A1805	04.06.2024	03.06.2029
k132	main meter	2814530	Jamde/220kv	0.2	ELSTER ALPHA A1800	04.06.2024	03.06.2029
	check meter	2814535	Jamde/220kv	0.2	ELSTER ALPHA A1800	04.06.2024	03.06.2029
J95	Main meter	2839063	Jamde/220kv	0.2	ELSTER ALPHA A1800	04.06.2024	03.06.2029
	check meter	2839062	Jamde/220kv	0.2	ELSTER ALPHA A1800	04.06.2024	03.06.2029
J96	Main meter	2839063	Jamde/220kv	0.2	ELSTER ALPHA A1800	04.06.2024	03.06.2029
	check meter	2839062	Jamde/220kv	0.2	ELSTER ALPHA A1800	04.06.2024	03.06.2029
J97	Main meter	2839063	Jamde/220kv	0.2	ELSTER ALPHA A1800	04.06.2024	03.06.2029
	check meter	2839062	Jamde/220kv	0.2	ELSTER ALPHA A1800	04.06.2024	03.06.2029
J98	Main meter	2839063	Jamde/220kv	0.2	ELSTER ALPHA A1800	04.06.2024	03.06.2029
	check meter	2839062	Jamde/220kv	0.2	ELSTER ALPHA A1800	04.06.2024	03.06.2029
G006	MAIN METER	14796421	Jamde/220kv	0.2	ELSTER ALPHA A1800	04.06.2024	03.06.2029
G-01	MAIN METER	7022930	Gadag 110/33kv	0.2	L & T	31.07.2024	31.06.2029
	CHECK METER	7022969	Gadag 110/33kv	0.2	L & T	31.07.2024	31.06.2029
G-03,G04,G08	MAIN METER	7022918	Gadag 110/33kv	0.2	L & T	31.07.2024	31.06.2029
	CHECK METER	7022904	Gadag 110/33kv	0.2	L & T	31.07.2024	31.06.2029
G-05	MAIN	7022956	Gadag	0.2	L & T	31.07.2024	31.06.2029

	METER		110/33kv				
	CHECK METER	7022946	Gadag 110/33kv	0.2	L & T	31.07.2024	31.06.2029
G-07	MAIN METER	7022960	Gadag 110/33kv	0.2	L & T	31.07.2024	31.06.2029
	CHECK METER	7022938	Gadag 110/33kv	0.2	L & T	31.07.2024	31.06.2029
G-10	MAIN METER	7022953	Gadag 110/33kv	0.2	L & T	31.07.2024	31.06.2029
	CHECK METER	7022923	Gadag 110/33kv	0.2	L & T	31.07.2024	31.06.2029
G-11	MAIN METER	22005279	Gadag 110/33kv	0.2	L & T	31.07.2024	31.06.2029
	CHECK METER	7022914	Gadag 110/33kv	0.2	L & T	31.07.2024	31.06.2029
G-12	MAIN METER	7022964	Gadag 110/33kv	0.2	L & T	31.07.2024	31.06.2029
	CHECK METER	7022941	Gadag 110/33kv	0.2	L & T	31.07.2024	31.06.2029
G-31	MAIN METER	7022948	Gadag 110/33kv	0.2	L & T	02.08.2024	02.07.2029
	CHECK METER	7022933	Gadag 110/33kv	0.2	L & T	02.08.2024	02.07.2029
G32,G33,G34	MAIN METER	7022961	Gadag 110/33kv	0.2	L & T	02.08.2024	02.07.2029
	CHECK METER	7022928	Gadag 110/33kv	0.2	L & T	02.08.2024	02.07.2029

There is calibration delay of Maharashtra turbine location no G06 for the monitoring period from 01/09/2020 to 31/12/2024. The error factor has been applied in net export values for the delayed period as meters were not calibrated as per the calibration frequency which is once in five years. As per the Appendix calibration of the VVS Standard v3.0, Para 366(a): error factor of " $\pm 0.2\%$ " should be applicable for both export & import i.e. the measured values. However, net electricity generation is considered as per the registered monitoring plan, the separate export and import values are not available. Hence being conservative and to account for the error for both export & import, a cumulative error of " -0.4% " on net electricity generation has been applied for delay period.³

³ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210921115831128/reg_stan06_v03.0.pdf?